Eco-Lies: Ecosystems, Food Webs and Pesticides

Who eats whom, where the energy goes, and what happens when poisons enter the food chain

1. Ecosystems

Ecology is the study of organisms, their environment and the interactions between them. It can be studied at many scales:

Level	Meaning	Example
Organism	One living thing	One mangrove tree
Population	All of one species in an area	All the tigers of a forest
Community	All the populations living together	Plants, animals, fungi and microbes of a pond
Ecosystem	The community plus the non-living environment	A pond with its water, light and sediments
Biome	A large region with a typical climate and life	Tropical rainforest, desert
Biosphere	All of Earth's life	The whole planet

An ecologist wants to understand the life processes, habitats, interactions and biodiversity of species.

Biotic and abiotic factors

- **Biotic** components are the living things: bacteria, plants, fungi, animals, birds and more.
- **Abiotic** components are the non-living chemical and physical parts: sunlight, soil, air, water, moisture, minerals and temperature. They belong to the lithosphere (land), hydrosphere (water) and atmosphere (air).

Two examples of ecology at work

- **Human ecology** studies the relationship between people and the environment — how we change it and how we can do better for people and nature.
- **Niche construction** studies how organisms change their environment for their own benefit and for others. Termites build mounds that can be several metres tall, which keep the colony cool, protect it and even grow food (fungus). Beavers build dams that create ponds for many other species.

QUICK ACTIVITY: SORT THE FACTORS (5 MINUTES)
Look at a picture of a pond. Make two lists, biotic and abiotic. Which list is longer? Which factors would change if the water were polluted?

2. Trophic levels

Every organism in an ecosystem is part of a food web. A food chain is one path that energy and nutrients take; all the overlapping food chains together form a food web. Organisms are sorted into trophic levels (feeding levels).

Level	Name	Examples	What they do
1	Producers (autotrophs)	Plants, algae, phytoplankton, some bacteria	Make their own food, usually by photosynthesis from sunlight, CO_2 and water
2	Primary consumers	Deer, grasshoppers, zooplankton	Herbivores — eat producers
3	Secondary consumers	Frog, snake eating a mouse, small fish	Carnivores that eat herbivores
4	Tertiary consumers	Owl or eagle eating a snake	Eat secondary consumers
top	Apex predators	Tiger, lion, great white shark, osprey	No natural enemies (except humans); at the 4th or 5th level
—	Decomposers & detritivores	Fungi, bacteria; earthworms, dung beetles	Break down dead matter and return nutrients to soil and water

A few details

- **Producers:** Plants are the most familiar, but algae (seaweeds are large algae), phytoplankton (tiny floating algae of the oceans) and some bacteria are producers too. Some bacteria near deep-sea vents and volcanoes use chemical energy from substances such as hydrogen sulphide instead of sunlight. This is **chemosynthesis**.
- **Omnivores,** like humans and bears, eat both plants and animals and so can sit at more than one level.
- **Scavengers** (such as vultures) eat the bodies of dead animals; **detritivores** (earthworms, dung beetles) eat dead plant and animal remains and droppings; **decomposers** (fungi and bacteria) finish the job, turning organic waste into the nutrients that producers use. That begins a whole new set of food chains.

3. Food chains and food webs

Food chains can be short or long, and many chains join into webs. Two chains from the same forest:

- grass → rabbit → fox → (when it dies) decomposers such as worms and fungi → nutrients back to the soil → grass

- tree leaves → caterpillar → sparrow → snake → eagle → vulture → soil bacteria

In the sea: algae and plankton → krill (tiny shrimp) → blue whale → orca. When a whale dies, its body sinks, and worms and bacteria on the sea floor release nutrients for new plankton.

GAME: GUESS THE FOOD CHAIN (15 MINUTES, RAPID FIRE)
Show a picture of an ecosystem. Students name one organism for each trophic level, one at a time. Do three ecosystems. Suggested: a grassland, a pond, and the Sundarbans mangrove. The answer tables below can be used as a key.

Level	Grassland	Pond	Sundarbans mangrove
Producer	Grass	Algae, water plants	Mangrove trees, phytoplankton; leaf litter feeds the web
Primary consumer	Grasshopper, deer	Tadpole, water snail	Crabs, zooplankton, mud-dwelling snails
Secondary consumer	Frog, mouse	Small fish, dragonfly nymph	Mudskippers, prawns, small fish
Tertiary / top	Snake → eagle	Large fish → heron	Large fish, fishing cat, saltwater crocodile, Bengal tiger
Decomposer	Fungi, soil bacteria	Bacteria, fungi	Fungi, bacteria in the mud

Energy and biomass

Biomass is the mass of living matter in organisms (and so the energy stored in it). Producers convert sunlight into biomass. At each step up, only about 10 % of the energy is passed on; the rest is used for life processes (movement, warmth, growth), lost as heat, or left uneaten. So biomass and numbers usually fall at each level: there are far more grass plants than grasshoppers, and far more grasshoppers than frogs.

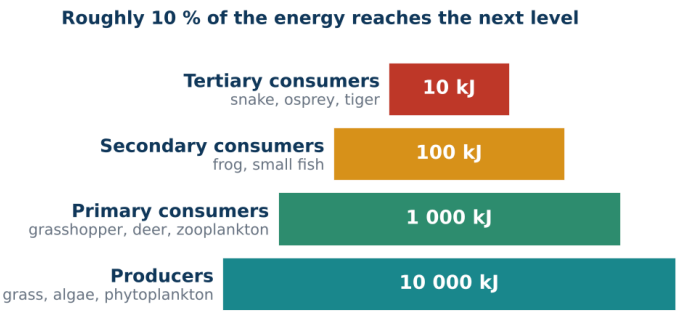


Figure 1. An energy pyramid: about 10 % of the energy is passed up each level. This is also why food chains rarely have more than four or five levels. *Source: Own illustration*

A healthy food web has many producers, fewer herbivores and relatively few carnivores. An ecosystem cannot support many top predators without a much larger base of herbivores and plants.

4. Biomagnification

Some poisons move up the food chain and get more concentrated at each level. This is biomagnification. Do not mix it up with bioaccumulation, which is the build-up of a substance in one organism over its lifetime.

A substance will biomagnify when it is:

- **persistent** — it does not break down quickly in the environment;
- **fat-soluble** — it dissolves in body fat and is stored there;
- **not removed** — the body cannot break it down or excrete it.

Examples are the pesticide DDT, industrial chemicals called PCBs, and heavy metals such as mercury (as methylmercury in fish). Small organisms take the poison in from water and food. A predator eats many contaminated prey, so it takes in the poison from all of them, and stores it. The animals at the top of the chain carry the largest amounts.

Case	What happened
DDT and fish-eating birds	DDT sprayed on land washed into water and built up in fish. Ospreys, eagles and pelicans that ate the fish laid eggs with thin shells that broke, and their numbers fell. Rachel Carson described this in her 1962 book *Silent Spring*.
Mercury in fish	In Minamata, Japan (from the 1950s), mercury waste from a factory got into fish, and people who ate the fish developed serious nerve damage.

FUN ACTIVITY
Do the Biomagnification hands-on activity (next document) to see the effect with sweets.

5. Pesticides and biopesticides

Pesticides are substances that kill, deter or control pests. The word includes insecticides (insects), herbicides (weeds), rodenticides (rodents) and fungicides (fungi, moulds and mildew). They are widely used to protect crops and livestock and to raise farm output.

The problem with chemical pesticides

- A large share of what is sprayed **never reaches the target**. It drifts in the air, runs off into water, settles in sediments, and ends up in our food.
- Pesticides are linked to health problems: short-term (headache, nausea) and long-term (cancer, harm to reproduction).
- They can harm non-target organisms — bees, earthworms, fish and birds — and reduce soil life, which lowers soil health and water retention.
- Pests can evolve **resistance**, so more and stronger chemicals are needed.

Biopesticides

Biopesticides are pesticides derived from natural materials: animals, plants, bacteria and certain minerals. For example, canola oil and baking soda have pest-controlling uses; in India, neem (azadirachtin) is a very common one.

Class	How it works	Example
Biochemical	Natural substances that control pests by non-toxic means — such as interfering with mating or attracting pests to traps	Insect sex pheromones; scented plant extracts in traps; neem oil
Microbial	A microorganism (bacterium, fungus, virus or protozoan) is the active ingredient. Each is fairly specific for its target.	*Bacillus thuringiensis* (Bt), which makes proteins that kill particular insect larvae; *Trichoderma* fungi
Plant-incorporated protectants (PIPs)	The plant is given a gene (for instance for the Bt protein) so that it makes the pesticide itself	Bt cotton — the only genetically modified crop approved for farming in India

How Bt works: each Bt strain makes a protein that binds to a receptor in the gut of certain larvae. This damages the gut, and the larva stops feeding and starves. Different strains target moths, flies or mosquitoes, so it is specific.

ADVANTAGES OF BIOPESTICIDES

- Usually less toxic than conventional pesticides.
- Usually affect only the target pest and close relatives (conventional pesticides often kill many other animals).
- Often work in small amounts and break down quickly, so there is less pollution.
- Used within **Integrated Pest Management (IPM)** — which combines biological, cultural and limited chemical controls — they can greatly cut the use of conventional pesticides while keeping yields high.

BUT...

To use biopesticides effectively and safely, farmers need to know a good deal about pest management and must follow label directions carefully. "Natural" does not automatically mean "harmless".

Check your understanding

1. Why do food chains rarely have more than five levels?
2. What is the difference between bioaccumulation and biomagnification?
3. Why is DDT more dangerous to an osprey than to a grasshopper?
4. Give one advantage and one limitation of biopesticides.
5. Put these in order from producer to apex predator: tiger, grass, chital deer.